

EE6211-POWER ELECTRONICS LAB

Experiment 3 Report

submitted by

Mohnish Sahu

EE25MTECH14018

Course Instructor:

Dr. RUPESH WANDHARE



Department of Electrical Engineering
Indian Institute of Technology, Hyderabad

Jan 2026

Contents

1	Experiment 1	1
1.1	Aim	1
1.2	Software and Components	1
1.3	Theory	2
1.4	Calculation of Components	2
1.4.1	Switch (MOSFET) Rating	2
1.4.2	Diode Rating	3
1.4.3	Parameter Values	4
1.4.4	Capacitor Selection	4
1.4.5	Inductor Non-Ideality Analysis	5
1.4.6	Transfer Function	5
1.4.7	Control-to-Output Transfer Function	5
1.4.8	PI Controller Design	5
1.4.9	Phase Margin	6
1.4.10	Simulation Results	6
1.4.11	No-Load Condition Observation	7
2	Learnings/Observations	8

1. Experiment 1

1.1 Aim

Simulation, complete modeling, design and control of a DC-DC boost converter with closed-loop control for application/rating presented in “Experiment-1”

- To design a DC–DC boost converter by selecting appropriate components
- To perform modeling and control design compared with open loop control for various output voltage references in MATLAB/Simulink simulations.
- Simulate boost converter in no load for few cycles and write your observations.

1.2 Software and Components

- MATLAB 2025a
- Resistor, Capacitor
- Voltage and current measurement blocks
- Scope blocks for waveform observation

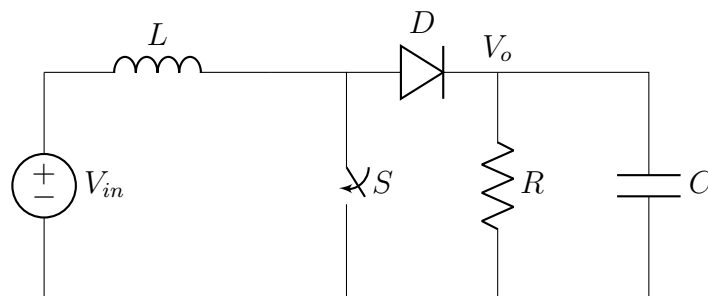


Figure 1.1: Boost Converter Circuit

1.3 Theory

A boost converter is a DC–DC power converter that increases the input voltage to a higher output voltage by storing energy in an inductor when the switch is ON and releasing it to the load when the switch is OFF. During the ON state, the inductor charges and the diode is reverse biased, while during the OFF state, the inductor discharges through the diode, adding to the input voltage and boosting the output.

1.4 Calculation of Components

Given Parameters

- $V_{in} = 48 \text{ to } 62V$
- $V_o = 64 \text{ to } 85V$
- $P_o = 200W$
- $f_s = 80kHz$
- Efficiency $\eta = 95\%$
- Inductor ripple $\Delta I_L = 20\%$
- Window utilization factor $K_w = 0.25$
- Current density $J = 3.5 A/mm^2$
- Maximum flux density $B_{max} = 0.25T$

1.4.1 Switch (MOSFET) Rating

Voltage Rating

In a boost converter, when the switch is OFF:

$$V_{switch} = V_o = 85V \quad (1.1)$$

Considering a safety margin of 1.5:

$$V_{rating} \geq 1.5 \times V_o = 1.5 \times 85 = 127.5V \quad (1.2)$$

Preferred Value: 150V MOSFET

Current Rating

The switch carries the inductor current:

$$I_{switch,avg} = I_{in} = 4.39A \quad (1.3)$$

Peak current:

$$I_{switch,peak} = I_{in} + \frac{\Delta I_L}{2} = 4.39 + \frac{0.878}{2} = 4.83A \quad (1.4)$$

Including safety margin:

$$I_{rating} \geq 1.5 \times 4.83 = 7.25A \quad (1.5)$$

Preferred Value: 10A MOSFET

1.4.2 Diode Rating

Voltage Rating

When the diode is reverse biased, it must block the output voltage:

$$V_D = V_o = 85V \quad (1.6)$$

With safety margin:

$$V_{rating} \geq 1.5 \times 85 = 127.5V \quad (1.7)$$

Selected Value: 150V diode

Current Rating

The diode carries the output current:

$$I_{D,avg} = I_o = \frac{P}{V_o} = \frac{200}{85} = 2.35A \quad (1.8)$$

Peak current:

$$I_{D,peak} = I_{L,peak} = 4.83A \quad (1.9)$$

With safety margin:

$$I_{rating} \geq 1.5 \times 4.83 = 7.25A \quad (1.10)$$

Selected Value: 10A diode

1.4.3 Parameter Values

From Experiment 1:

$$L = 297.5 \mu H \quad (1.11)$$

Capacitance:

$$C = \frac{I_o d}{f_s \Delta V_o} \quad (1.12)$$

Duty cycle limits:

$$D_{max} = 1 - \frac{V_{in}}{V_o} = 1 - \frac{48}{85} = 0.435 \quad (1.13)$$

$$D_{min} = 1 - \frac{48}{64} = 0.25 \quad (1.14)$$

Allowed ripple:

$$\Delta V_o = 0.03 V_o = 85 \times 0.03 = 2.55 V \quad (1.15)$$

Output current range:

$$I_o = 2.35 \text{ to } 3.125 A \quad (1.16)$$

1.4.4 Capacitor Selection

$$\Delta V_o = \frac{I_o D}{f_s C} \quad (1.17)$$

$$C = \frac{I_o D}{f_s \Delta V_o} \quad (1.18)$$

$$\Delta V_o = 0.01 \times 85 = 0.85 V \quad (1.19)$$

$$C = \frac{2.48 \times 0.435}{80 \times 10^3 \times 0.85} \approx 15.85 \mu F \quad (1.20)$$

$$C = 22 \mu F \quad (1.21)$$

—

1.4.5 Inductor Non-Ideality Analysis

Switch ON:

$$V_L = V_{in} - i_L r_L \quad (1.22)$$

Switch OFF:

$$V_L = V_{in} - i_L r_L - V_o \quad (1.23)$$

Volt-second balance:

$$(V_{in} - i_L r_L)D + (V_{in} - i_L r_L - V_o)(1 - D) = 0 \quad (1.24)$$

$$V_o = \frac{V_{in} - i_L r_L}{1 - D} \quad (1.25)$$

1.4.6 Transfer Function

$$\frac{V_o(s)}{V_{in}(s)} = \frac{1 - D}{LCs^2 + \frac{s}{RC} + \frac{(1-D)^2}{LC}} \quad (1.26)$$

$$\frac{V_o(s)}{V_{in}(s)} = \frac{1.79 \times 10^8}{s^2 + 2768.2s + 1.1 \times 10^8} \quad (1.27)$$

1.4.7 Control-to-Output Transfer Function

$$G_{vd}(s) = \frac{V_g}{(1 - D)^2} \cdot \frac{1 - \frac{sL}{R(1-D)^2}}{LCs^2 + \frac{L}{R}s + (1 - D)^2} \quad (1.28)$$

$$G_{vd}(s) = 150.33 \cdot \frac{1 - 2.58 \times 10^{-5}s}{6.545 \times 10^{-9}s^2 + 8.23 \times 10^{-6}s + 0.319225} \quad (1.29)$$

1.4.8 PI Controller Design

$$G_c(s) = K_p + \frac{K_i}{s} \quad (1.30)$$

Crossover Frequency

$$f_z \approx 6.16kHz \quad (1.31)$$

$$f_c \leq \frac{f_z}{5} \approx 1.2kHz \quad (1.32)$$

$$f_c = 800Hz, \quad \omega_c = 5026rad/s \quad (1.33)$$

PI Zero

$$f_i = 160Hz \quad (1.34)$$

$$\omega_i = 1005rad/s \quad (1.35)$$

—

Calculation of K_p

$$|G_c(j\omega_c)G_{vd}(j\omega_c)| = 1 \quad (1.36)$$

$$|G_{vd}(j\omega_c)| \approx 939 \quad (1.37)$$

$$K_p = \frac{1}{939} \approx 0.00106 \quad (1.38)$$

—

Calculation of K_i

$$K_i = K_p \cdot \omega_i = 0.00106 \times 1005 \approx 1.065 \quad (1.39)$$

—

Final Controller

$$G_c(s) = 0.00106 + \frac{1.065}{s} \quad (1.40)$$

—

1.4.9 Phase Margin

$$PM = 180^\circ + \angle G_{OL}(j\omega_c) \quad (1.41)$$

$$\angle G_c = 78.7^\circ \quad (1.42)$$

$$\angle G_{vd} = -22.4^\circ \quad (1.43)$$

$$PM = 78.7^\circ - 22.4^\circ = 56^\circ \quad (1.44)$$

1.4.10 Simulation Results

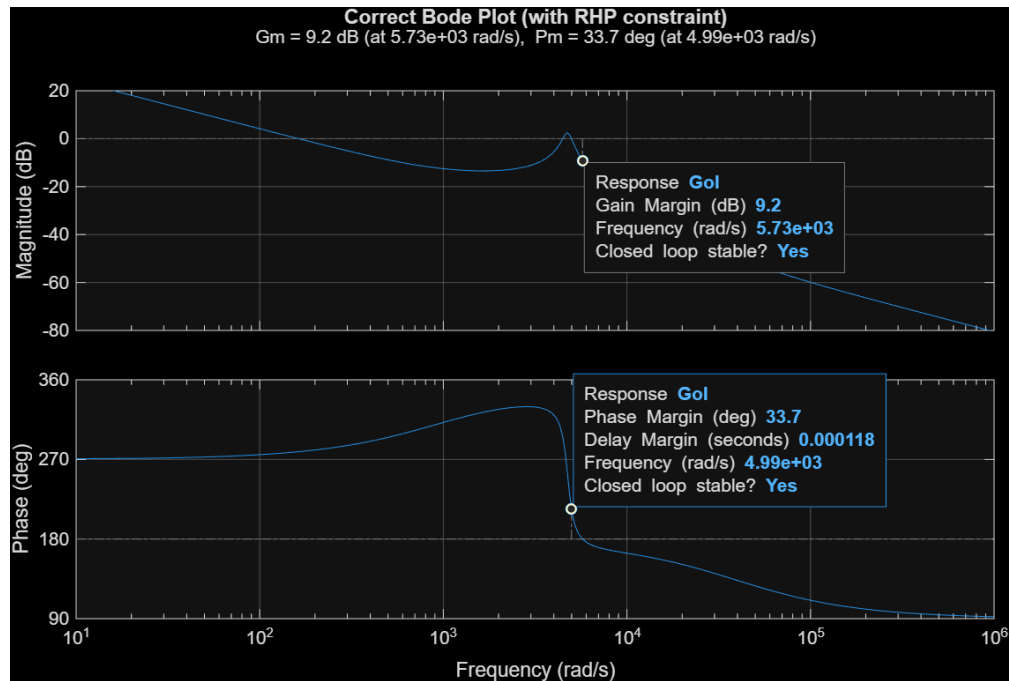


Figure 1.2: Bode Plot

1.4.11 No-Load Condition Observation

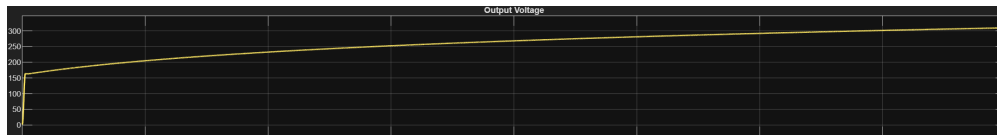


Figure 1.3: Voltage waveform for No load

1. When a boost converter is simulated under no-load (open circuit) for a few switching cycles, the output capacitor has no discharge path and continuously accumulates energy. As a result, the output voltage keeps increasing and may reach very high values instead of stabilizing.
2. Although the inductor continues its normal charging and discharging operation, the stored energy is entirely transferred to the capacitor due to the absence of load.
3. This leads to voltage overshoot and unstable behavior. Hence, a boost converter cannot regulate its output voltage properly without a load or appropriate feedback control, and a minimum load is required for stable operation.

2. Learnings/Observations

1. The actual voltage gain is slightly lower than the ideal gain due to the presence of inductor parasitic resistance. The effect is minimal because the inductor resistance is very small. However, for higher resistance values, the reduction in output voltage becomes significant.
2. The system also exhibits a right-half plane zero, limiting the control bandwidth.
3. During transient conditions, the output takes time to reach steady state and may show overshoot, which is improved using closed-loop control.